Entropy & Information

Fill-in study notes on Shannon's theory

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dyed in indigo, madder & weld ruled in iron-gall.   woven on loom.

1 INFORMATION CONTENT

A random variable X with outcomes $x_1, \ldots, x_n$ and probabilities $p_1, \ldots, p_n$.

↝ entropy

Definition 1.1 *(Shannon, 1948).* The **entropy** of X is defined as

$$H(X) = -\sum_{i=1}^{n} p_i \log_2 p_i$$

measured in **bits**.

> Think of entropy as "average surprise." If you already know the outcome, surprise is zero. If every outcome is equally likely, surprise is maximised.

The information content of a single outcome x_i is ________________.

? Why $\log_2$ and not $\ln$?

Theorem 1.2 *(Maximum entropy).* Among all distributions on n outcomes, entropy is maximised by the ______________ distribution, achieving $H = \log_2 n$ bits.

Proof. Apply Jensen's inequality to the concave function $f(p) = -p \log p$.
[fill in: complete the Jensen argument]

1.1 Properties of Entropy

Key identities

Trigger. When comparing two sources, reach for relative entropy.

1. $H(X) \geq 0$ with equality iff X is deterministic ■■■■■
2. $H(X) \leq \log_2 |\mathcal{X}|$ with equality iff X is uniform ■■■■□
3. $H(X,Y) = H(X) + H(Y|X)$ (chain rule) ■■■□□

What's the chain rule for three variables?

Your turn. Compute $H(X)$ for a biased coin with $p = \frac{1}{4}$:

1.2 The Identities (the information diagram)

identity	reading
$H(X,Y) = H(X) + H(Y\|X)$	**chain rule**: total = first + rest
$I(X;Y) = H(X) + H(Y) - H(X,Y)$	shared = overlap of the two circles
$I(X;Y) = D_{\mathrm{KL}}(p_{XY} \parallel p_X p_Y)$	information = distance from **independence**
$I(X;Y) \geq 0$, with $= 0$ iff independent	you never lose by looking at Y

2 MUTUAL INFORMATION

↝ mi

Definition 2.1. The **mutual information** between X and Y is

$$I(X;Y) = H(X) - H(X|Y) = H(Y) - H(Y|X)$$

Mutual information measures how much knowing Y tells you about X. It's symmetric: knowing X tells you the same amount about Y.

Example 2.2. Let X be a fair coin and $Y = X$. Then $I(X;Y) = H(X) = 1$ bit. The copy gives you everything. [entropy ↩]

Lemma 2.3 *(Non-negativity).* $I(X;Y) \geq 0$ with equality iff X and Y are independent.

Remark (Connection to KL divergence). $I(X;Y) = D_{\mathrm{KL}}(p_{X,Y} \parallel p_X p_Y)$, which makes non-negativity immediate from Gibbs' inequality.

L03

Corollary 2.4. Data processing inequality: if $X \to Y \to Z$ forms a Markov chain, then $I(X;Z) \leq I(X;Y)$.

Proof. Expand using chain rule for mutual information. The Markov condition $p(z|x,y) = p(z|y)$ kills the extra term. ∎